

Four-H100 training improvements

26–27 September 2026

90.6 iterations/s sustained live training +25.7%

Versus 72.1 it/s with the corrected loader and starting GPU kernels. Four H100 SXM 80 GB; global batch 131,072; FP32; AdamW; no-PSQT threats network, L1=1024 and L2=L3=32. All production filters remain enabled, including check/capture filtering and random FEN skipping of 10.

Measured progression

Mature-network configuration	Live it/s	Gain over baseline
Corrected loader, starting GPU kernels	72.10	—
FT launch/atomic tuning + bucketed dense L1 + DDP	85.39	18.4%
Add weighted-NNZ L1 forward	86.40	19.8%
Add eight-position FT gradient aggregation	90.62	25.7%

Medians of three 2,048-step windows after 1,024 warmup steps. Final cached throughput: 93.76 it/s.

Retained improvements

Loader allocation reuse	glibc arena/mmap/trim settings reduce native batch release from 10.5 to 0.024 ms.
Loader staging threads	Four torch threads and 64 loader workers per rank; combined loader tuning raises roughly 37 to 72 it/s. Production sampling is preserved.
BMI2 detection	Match the complete CPU flag so AVX-512 VBMI2 does not disable the decoder fast path. No isolated speedup attributed.
Loss metric updates	Remove per-update GPU-to-CPU NaN checks; retain nonfinite-loss checks at logging boundaries. Included in the corrected baseline.
FT forward launch	256-thread blocks for H100/L1=1024: about 1.69 to 1.50 ms. General shapes retain their existing path.
FT backward launch	Four 128-thread column blocks plus zero-gradient atomic guards: about 5.34 to 3.84 ms. These are separate commits.
FT feature aggregation	Combine updates across eight positions before global atomics. About 3.18 ms backward + 0.18 ms packing; +4.9% live throughput. Duplicate indices use a correctness fallback.
Selected-bucket L1	Compute the selected bucket directly. Input-gradient kernel: 0.381 to 0.104 ms; weight-gradient kernel: 0.330 to 0.107 ms, plus small reductions.
Weighted-NNZ L1	Compact nonzeros and coalesce weight reads. Forward: 0.315 to 0.166 ms including packing/transpose; about +1.3% whole-step throughput at 25% activation density.
DDP bucket size	50 to 400 MiB reduces gradient collectives from three to one. Pre-aggregation four-GPU scaling was 3.18× (79.6% efficiency); scaling is not proven optimal.

Hardware-counter finding

A full 5090 VM enabled Nsight Compute counters. With cache flushing disabled, FT forward reached **94% L2 / 26% DRAM** throughput; direct backward reached **73% L2 / 12% DRAM**. This points to cache/atomic pressure on that GPU. Aggregation reduced L2 reduction sectors by 17.7% in the default-cache run. These are **5090 diagnostics, not H100 bandwidth measurements**.

Validation and limits

104 tests passed, 2 skipped on H100; full production-size validation and checkpoint resume through step 2,560 completed with finite loss. The mature net came from the supplied public `nn-252f33942263.nnue`; comparisons used identical fresh optimizer state. Sparse L1 can regress dense early nets by about 1%. Quantization fusion, separate bias reductions, and forced NCCL protocols did not yield useful whole-training gains. No strength claim is made. After rebase, each branch passes 47 local CPU tests (59 skipped). GPU timings predate the history rewrite; measured runtime sources are preserved.

Branches and evidence

`optimize/4xh100-no-loader` GPU/DDP work, baseline loader launch settings.
`optimize/4xh100-with-loader` Adds separate BMI2, allocation-reuse, and staging-thread commits.
Both are based on master `aa2fff2`; upstream HyperLogLog changes remain in both. Added Markdown reports remain untracked and unstaged.

Evidence: `docs/benchmarks/h100-followup-2026-09-26/`.

Workload: [nettest threats.yaml](#). Source: `h100-improvements.tex`.